AP PRECALCULUS

Unit 1 • Course notes

Polynomial and Rational Functions

Topics 1.1 to 1.14 • About 30–40% of the AP Precalculus exam

In this unit you learn to describe how the outputs of a function change with its inputs, to analyze polynomial and rational functions from their formulas, graphs and tables, and to build function models from data and contexts.

1.1 Change in Tandem

A **function** assigns exactly one output value to each input value. As the input values change, the output values change *in tandem*.

Key idea

- f is **increasing** on an interval if, as inputs increase, outputs increase: $a < b \Rightarrow f(a) < f(b)$.
- f is **decreasing** on an interval if, as inputs increase, outputs decrease: $a < b \Rightarrow f(a) > f(b)$.
- The graph is **concave up** where the rate of change is increasing, and **concave down** where the rate of change is decreasing.
- A **zero** of f is an input a with $f(a) = 0$; it is an x-intercept of the graph.

Example

A cup of coffee cools in a room. Its temperature T (in °C) is recorded every 5 minutes.

t (min)	0	5	10	15	20
$T(t)$ (°C)	90	70	56	46	40

Solution. The outputs decrease, so T is decreasing. The changes are $-20, -14, -10, -6$: the temperature falls more and more slowly. The rate of change is negative and increasing, so the graph is decreasing and concave up.

1.2 Rates of Change

Key idea

The **average rate of change** of f on the interval $[a, b]$ is

$$\frac{f(b) - f(a)}{b - a}.$$

It is the slope of the **secant line** through $(a, f(a))$ and $(b, f(b))$. Its units are *output units per input unit*.

The **rate of change at a point** (instantaneous rate of change) can be approximated by average rates of change over smaller and smaller intervals that contain the point.

Example

Let $f(x) = x^2$. Find the average rate of change on $[1, 3]$ and estimate the rate of change at $x = 1$.

Solution. On $[1,3]$: $\dfrac{f(3)-f(1)}{3-1} = \dfrac{9-1}{2} = 4$.
On $[1,\ 1.01]$: $\dfrac{1.0201-1}{0.01} = 2.01$. The rate of change at $x=1$ is about 2.

Sign of the rate of change
Where the rate of change is positive, f is increasing; where it is negative, f is decreasing. Where f changes from increasing to decreasing (or the reverse), the rate of change changes sign.

1.3 Rates of Change in Linear and Quadratic Functions

Key idea
- A **linear** function has a *constant* average rate of change on every interval: its slope.
- For a **quadratic** function, the average rates of change over consecutive equal-length intervals change at a constant rate. In a table with equally spaced inputs, the *first differences* change by a constant amount: the *second differences* are constant.

Example

x	0	1	2	3	4
y	1	4	9	16	25
1st differences		3	5	7	9
2nd differences			2	2	2

Solution. The second differences are constant, so the data can be modeled by a quadratic function. Here $y = (x+1)^2$.

For a quadratic $f(x) = ax^2 + bx + c$: if $a > 0$ the rate of change increases (concave up, minimum at the vertex); if $a < 0$ it decreases (concave down, maximum at the vertex).

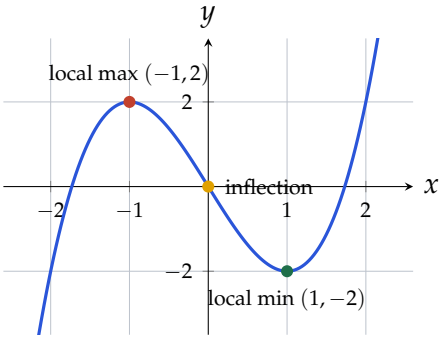
1.4 Polynomial Functions and Rates of Change

A **polynomial function** of degree n has the form

$$p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0, \qquad a_n \neq 0.$$

a_n is the **leading coefficient** and $a_n x^n$ the **leading term**.

Key idea
- A **local (relative) maximum** occurs where p changes from increasing to decreasing; a **local minimum** where it changes from decreasing to increasing.
- The greatest local maximum on the whole domain is the **global (absolute) maximum**; similarly for the minimum.
- A polynomial of degree n has at most $n-1$ local extrema.
- A **point of inflection** is where the graph changes concavity: the rate of change goes from increasing to decreasing, or the reverse.



$p(x) = x^3 - 3x$: increasing on $(-\infty, -1)$ and $(1, \infty)$, decreasing on $(-1, 1)$, concave down on $(-\infty, 0)$, concave up on $(0, \infty)$.

1.5 Polynomial Functions and Complex Zeros

Key idea

- If a is a real zero of p, then $(x - a)$ is a factor of $p(x)$, and $(a, 0)$ is an x-intercept.
- A polynomial of degree n has exactly n complex zeros, counted with **multiplicity**.
- If p has real coefficients, its non-real zeros come in **conjugate pairs**: if $a + bi$ is a zero, so is $a - bi$.
- At a real zero of **even** multiplicity, the graph *touches* the x-axis and turns back. At a zero of **odd** multiplicity, it *crosses* the x-axis (flattening out if the multiplicity is 3 or more).

Example

Describe the zeros of $p(x) = (x - 2)^2(x + 1)(x^2 + 4)$.
Solution. p has degree $2 + 1 + 2 = 5$, so it has 5 complex zeros: $x = 2$ (multiplicity 2, the graph touches the axis), $x = -1$ (multiplicity 1, the graph crosses), and $x = \pm 2i$ (from $x^2 + 4 = 0$), which are not x-intercepts.

Even and odd functions
f is **even** if $f(-x) = f(x)$ for all x (graph symmetric about the y-axis). f is **odd** if $f(-x) = -f(x)$ (graph symmetric about the origin).

1.6 Polynomial Functions and End Behavior

As $x \to \pm\infty$, a polynomial behaves like its leading term $a_n x^n$.

	$a_n > 0$	$a_n < 0$
n even	$\lim_{x \to \pm\infty} p(x) = +\infty$	$\lim_{x \to \pm\infty} p(x) = -\infty$
n odd	$\lim_{x \to -\infty} p(x) = -\infty$, $\lim_{x \to \infty} p(x) = +\infty$	$\lim_{x \to -\infty} p(x) = +\infty$, $\lim_{x \to \infty} p(x) = -\infty$

Example

Find the end behavior of $p(x) = (1 - x)(x + 2)^3$.
Solution. The leading term is $(-x)(x^3) = -x^4$: even degree, negative leading coefficient. So $\lim_{x \to -\infty} p(x) = -\infty$ and $\lim_{x \to \infty} p(x) = -\infty$.

1.7 Rational Functions and End Behavior

A **rational function** is a quotient of polynomials, $r(x) = \dfrac{p(x)}{q(x)}$ with q not the zero polynomial. Its end behavior is that of the quotient of the leading terms.

Key idea

Let $\deg p = m$ and $\deg q = n$.

- $m < n$: horizontal asymptote $y = 0$.
- $m = n$: horizontal asymptote $y = \dfrac{\text{leading coefficient of } p}{\text{leading coefficient of } q}$.
- $m = n + 1$: slant asymptote $y =$ quotient of the polynomial division $p \div q$.
- $m > n + 1$: no linear asymptote; r behaves like a polynomial of degree $m - n$.

Example

$r(x) = \dfrac{6x^2 - 1}{2x^2 + x}$ has horizontal asymptote $y = \dfrac{6}{2} = 3$, so $\lim_{x \to \pm\infty} r(x) = 3$.

1.8 Rational Functions and Zeros

Key idea

The zeros of $r = \dfrac{p}{q}$ are the real zeros of the numerator p that are *in the domain* of r (that is, not zeros of q).

Example

$r(x) = \dfrac{x^2 - 4}{x + 1}$ has zeros $x = 2$ and $x = -2$. The function $s(x) = \dfrac{x^2 - 4}{x - 2}$ has only one zero, $x = -2$: $x = 2$ is not in its domain.

The sign of a rational function can only change at its zeros and at the values excluded from its domain. A sign chart on the intervals between these values solves inequalities such as $r(x) \geq 0$.

1.9 Rational Functions and Vertical Asymptotes

Key idea

If a is a zero of the denominator and, after factoring, the factor $(x - a)$ appears more times in the denominator than in the numerator, then the line $x = a$ is a **vertical asymptote**: $r(x) \to +\infty$ or $-\infty$ as $x \to a^-$ and as $x \to a^+$.

To decide between $+\infty$ and $-\infty$, test the sign of r just to the left and just to the right of a.

1.10 Rational Functions and Holes

Key idea

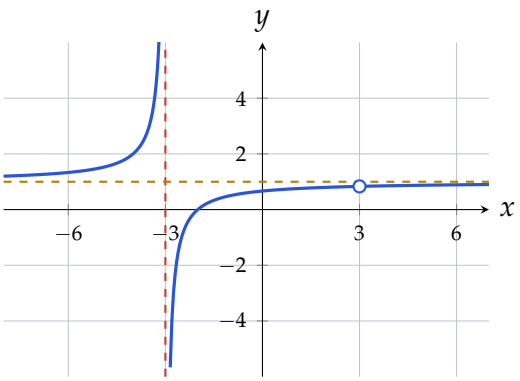
If $(x - a)$ appears in the numerator at least as many times as in the denominator, the graph has a **hole** at $x = a$. Cancel the common factors; the hole is at $(a,$ value of the simplified expression at $a)$, and $\lim_{x \to a} r(x)$ equals that value.

Example

Analyze $r(x) = \dfrac{x^2 - x - 6}{x^2 - 9}$.

Solution. Factor: $r(x) = \dfrac{(x-3)(x+2)}{(x-3)(x+3)}$.

- Hole at $x = 3$: $\dfrac{3+2}{3+3} = \dfrac{5}{6}$, so the hole is at $\left(3, \frac{5}{6}\right)$.
- Vertical asymptote $x = -3$: $\lim_{x \to -3^-} r(x) = +\infty$, $\lim_{x \to -3^+} r(x) = -\infty$.
- Zero: $x = -2$. Horizontal asymptote: $y = 1$.



AP exam tip

On the exam, justify every asymptote and hole with the factored form *and* a limit statement, for example "$\lim_{x \to -3^+} r(x) = -\infty$, so $x = -3$ is a vertical asymptote."

1.11 Equivalent Representations of Polynomial and Rational Expressions

The same function can be written in several forms; each reveals different information.

- **Standard form** $ax^2 + bx + c$: the y-intercept and end behavior.
- **Factored form** $a(x - r_1)(x - r_2)$: the zeros.
- **Quotient form** from polynomial division $p(x) = d(x)\,q(x) + r(x)$: slant asymptotes and end behavior of rational functions.

Polynomial division

Write $\dfrac{x^3 + 2x^2 - 5x + 1}{x - 1}$ as a quotient plus a remainder.

Solution. Synthetic division by 1 on the coefficients $1,\ 2,\ -5,\ 1$ gives $1,\ 3,\ -2$ and remainder -1:

$$\frac{x^3 + 2x^2 - 5x + 1}{x - 1} = x^2 + 3x - 2 - \frac{1}{x - 1}.$$

Binomial theorem

The coefficients of $(a + b)^n$ are the numbers in row n of Pascal's triangle. For example, $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$, so $(x + 2)^3 = x^3 + 6x^2 + 12x + 8$.

1.12 Transformations of Functions

Key idea

For $g(x) = a\,f\big(b(x-h)\big) + k$:

- k: vertical translation by k units; h: horizontal translation by h units.
- $|a|$: vertical dilation by a factor $|a|$; $a < 0$ adds a reflection over the x-axis.
- $\left|\frac{1}{b}\right|$: horizontal dilation by a factor $\frac{1}{|b|}$; $b < 0$ adds a reflection over the y-axis.

Horizontal changes affect the *domain*; vertical changes affect the *range*.

Example

f has domain $[-2, 4]$ and range $[0, 6]$. Find the domain and range of $g(x) = 3f(x-1) - 2$.
Solution. Domain: shift right 1, so $[-1, 5]$. Range: multiply by 3 then subtract 2, so $[3 \cdot 0 - 2,\ 3 \cdot 6 - 2] = [-2, 16]$.

1.13 Function Model Selection and Assumption Articulation

Choosing a model

- **Linear**: the data show a roughly constant rate of change.
- **Quadratic**: roughly constant second differences, or a single maximum or minimum.
- **Cubic / higher polynomial**: several turning points or a change in concavity.
- **Rational**: a quantity that approaches a limiting value, or an inverse relationship (for example, cost per unit $= \frac{\text{total cost}}{\text{units}}$).
- **Piecewise**: different behavior on different intervals.

A model relies on **assumptions** (the pattern continues, the context allows only some inputs). State them, and restrict the domain and range to values that make sense in context.

1.14 Function Model Construction and Application

A model can be built from a geometric or verbal description, or from data with technology (linear, quadratic, cubic or quartic regression).

Example

A box without a lid is made from a 20 cm by 30 cm sheet by cutting a square of side x cm from each corner and folding up the sides.
Solution. Volume: $V(x) = x(30 - 2x)(20 - 2x)$ cm^3. The side lengths must be positive, so the domain is $0 < x < 10$. For example, $V(3) = 3 \cdot 24 \cdot 14 = 1008$ cm^3. A table or graph shows the maximum volume is about 1056 cm^3, reached when $x \approx 3.9$ cm.

AP exam tip

When you use a model, give units and interpret your answer in context: "When the squares have side 3 cm, the box holds 1008 cm^3."

Unit 1 summary

- Average rate of change $= \dfrac{f(b) - f(a)}{b - a}$.
- Linear: constant rate of change. Quadratic: constant second differences.
- Degree n: n complex zeros, at most $n - 1$ turning points.
- Even multiplicity: touches. Odd: crosses.

- End behavior of a polynomial: its leading term.
- Rational end behavior: compare degrees.
- Zero of the denominator not cancelled: vertical asymptote.
- Common factor that cancels: hole.
- $a\,f(b(x-h))+k$: vertical changes act on the range, horizontal ones on the domain.
- Models: state assumptions, units, and a sensible domain.